

Title: Few-Shot Anomaly Detection in Time-Series Data Using a Hybrid Deep Learning Approach

1. Introduction

In today's data-driven world, large volumes of time-series data are continuously generated from various systems and devices, such as stock market prices, industrial sensors, server logs, website traffic, and healthcare monitoring systems. Detecting abnormal patterns in such data is critically important, as anomalies often indicate system failures, cyber-attacks, fraudulent activities, or mechanical faults. Early detection of these irregularities can prevent severe damage, financial loss, and operational disruptions.

However, a major challenge in anomaly detection is the scarcity of labeled anomaly data. In many real-world applications, abnormal events are rare, and collecting labeled datasets is costly and time-consuming. Traditional machine learning and deep learning methods typically require large amounts of labeled data to perform effectively. This limitation makes them less practical in scenarios where data is limited.

Few-Shot Learning offers a promising solution to this problem. Few-shot anomaly detection allows models to learn normal behavior from a very small number of samples and identify deviations without requiring extensive labeled datasets. This research proposes a hybrid deep learning framework that combines self-supervised learning and few-shot learning techniques to improve anomaly detection performance in time-series data with limited supervision.

2. Research Objectives

The main objectives of this research are:

- To develop an effective anomaly detection model for time-series data using limited labeled samples.
- To integrate self-supervised learning and few-shot learning into a unified framework.
- To design a hybrid architecture combining LSTM, Transformer, and Siamese Networks.
- To enhance the detection of unseen or open-set anomalies.
- To incorporate explainability mechanisms to improve model transparency and interpretability.

3. Proposed Methodology

The research will follow a structured methodological framework consisting of the following stages:

Step 1: Data Collection and Preprocessing

Time-series datasets from industrial or benchmark sources (e.g., HVAC systems or other real-world datasets) will be collected. Data preprocessing techniques such as normalization, noise filtering, missing value handling, and segmentation will be applied to prepare the data for training.

Step 2: Self-Supervised Pretraining

To overcome limited labeled data, self-supervised learning techniques will be applied. Methods such as Masked Time-Series Modeling and Contrastive Learning will be used to learn meaningful feature representations from unlabeled data. This stage helps the model understand general temporal patterns before supervised fine-tuning.

Step 3: Hybrid Model Development

A hybrid deep learning architecture will be designed, consisting of:

- LSTM (Long Short-Term Memory): To capture temporal dependencies in sequential data.
- Transformer: To model long-range dependencies and global contextual relationships.
- Siamese Network: To enable few-shot similarity-based learning using limited normal samples.
- Distance-Based Anomaly Scoring: To measure deviation from learned normal patterns.

This integrated framework aims to improve detection accuracy and generalization ability under low-data conditions.

Step 4: Model Evaluation

The proposed model will be evaluated using performance metrics such as:

- Area Under the ROC Curve (AUC)
- F1-Score
- Balanced Accuracy
- True Positive Rate (TPR) and False Positive Rate (FPR)

Comparative experiments will be conducted against traditional and state-of-the-art anomaly detection methods.

4. Expected Outcomes

- Improved anomaly detection performance with limited labeled data.
- Higher AUC and F1-scores compared to conventional approaches.
- Effective detection of unseen (open-set) anomalies.
- A scalable and adaptable model suitable for real-time monitoring systems.
- Improved interpretability through attention-based or explainability mechanisms.

5. Significance of the Study

Few-shot anomaly detection has significant practical importance in various domains, including industrial automation, healthcare monitoring, cybersecurity, and financial fraud detection. In many cases, small irregularities appear before major system failures occur. Detecting these early signs can help prevent large-scale damage and ensure system reliability.

This research contributes to the development of efficient anomaly detection systems that operate effectively in data-scarce environments. By integrating hybrid deep learning techniques and self-supervised pretraining, the proposed approach aims to bridge the gap between theoretical research and practical industrial applications.

6. Conclusion

Anomaly detection in time-series data remains a challenging problem, especially when labeled data is limited. By combining few-shot learning, self-supervised learning, and hybrid deep learning architectures, it is possible to design a robust and adaptable anomaly detection system. This research has the potential to provide meaningful contributions to both academic research and real-world applications, paving the way for more intelligent and efficient monitoring systems.

